FULL SIMULATION SET 03

UPSC Civil Services (Main) Simulation GS-III

DETAILED SOLUTIONS / ANSWER KEY | Time: 3 Hours | Maximum Marks: 250

Q1. Distinguish the money market from the capital market and explain why a developed economy needs both.

[10 marks | 150 words]

Model answer (144 words; practice model, not official): Money and capital markets differ chiefly in financing horizon and function. The money market redistributes short-term liquidity through call money, repo, Treasury Bills, Commercial Paper and Certificates of Deposit. It helps banks and firms bridge cash-flow mismatches and transmits monetary conditions. The capital market mobilises medium- and long-term finance through equity, dated Government securities, State Development Loans and corporate bonds. The distinction is not absolute safety versus risk. A short instrument can default or face rollover stress, while a long sovereign bond can be credit-safe yet price-sensitive. Both primary issuance and secondary liquidity matter: issuance transfers funds to the borrower; trading supplies price discovery and exit. A developed economy needs an integrated ladder. Money markets stabilise working liquidity, whereas capital markets fund infrastructure, enterprise and public borrowing. Transparent benchmarks and resilient settlement must connect them without allowing leverage or maturity mismatch to transmit shocks.

Q2. Distinguish A2, A2+FL and C2 and explain the official MSP formula.

[10 marks | 150 words]

Model answer (116 words; practice model, not official): A2 records paid-out cultivation expenses such as seed, fertiliser, hired labour, machinery and irrigation under the CACP method. A2+FL adds imputed unpaid family labour. C2 adds imputed rent on owned land and interest on owned fixed capital. The Union policy announced through the 2018-19 Budget framework fixes MSP at least 1.5 times the all-India weighted average A2+FL cost. CACP also considers C2 and wider demand, supply, parity and price factors. Thus the official 50 per cent margin claim and the farmer demand for C2 plus 50 per cent use different denominators. Neither average guarantees the same margin in practice on every farm because costs, yields and procurement access vary by crop and State.

Q3. The Gati-Shakti Yojana needs meticulous coordination between the government and the private sector to achieve the goal of connectivity. Discuss.

[10 marks | 150 words]

Model answer (138 words; practice model, not official): PM GatiShakti is a GIS-enabled planning framework that overlays infrastructure and socio-economic data to improve multimodal and last-mile connectivity. Coordination is necessary because the Union controls major networks and funding, States handle land and many clearances, local bodies provide urban services, and private firms contribute design, construction, operation, finance and logistics information. Shared route and utility data can prevent duplication, reveal missing links and sequence road, rail, port and industrial projects. The platform alone cannot deliver connectivity. Data quality, secure access, accountable ownership of each interface, timely land and environmental processes, interoperable standards and bankable procurement remain essential. Private inputs must not become opaque project selection or exclusive data advantage. Thus GatiShakti should be judged through reduced conflicts, faster but lawful preparation, integrated terminals, reliable last-mile service and lower lifecycle cost-not by layers uploaded or projects evaluated alone.

Q4. Distinguish delayed-payment remedies, TReDS, factoring and credit guarantees.

[10 marks | 150 words]

Model answer (126 words; practice model, not official): Delayed payments create an MSME liquidity crisis. The MSMED Act limits the agreed payment period for micro and small suppliers to forty-five days and provides statutory interest and MSEFC remedies. Finance instruments must match the problem. TReDS discounts accepted receivables through competing financiers and, under RBI's 23 June 2026 Directions, transactions are without recourse to sellers. Factoring assigns the payment claim; it is not machinery finance. CGTMSE instead shares lender loss on eligible credit, while SIDBI provides development and financing support. These mechanisms fail when buyers do not accept invoices, suppliers fear commercial retaliation, or enforcement remains slow. ODR can lower dispute cost but cannot erase bargaining power. Policy should track acceptance time, discount cost, recovery, repeat orders and cash-conversion days rather than portal registrations alone.

Q5. Distinguish statutory and economic tax incidence and explain why the distinction matters for equity.

[10 marks | 150 words]

Model answer (122 words; practice model, not official): Tax incidence has two layers. Statutory incidence identifies who must remit a tax; economic incidence identifies whose real income falls after prices, wages and returns adjust. A GST-registered seller remits tax, but consumers, workers or owners bear it according to elasticities, market power and adjustment time. This distinction changes equity analysis. A consumption tax may absorb more of a poor household's income, while a corporate levy may partly shift through wages or prices. Transfers can offset or reinforce the burden. Therefore, progressivity cannot be inferred from the statutory rate or taxpayer name. Analysis must identify the legal base, trace shifting and assess the tax-and-transfer outcome. However, incidence estimates vary by market structure and time horizon; they are evidence-based distributions, not fixed labels.

Q6. Why is the phosphorus cycle analytically different from carbon and nitrogen cycles?

[10 marks | 150 words]

Model answer (96 words; practice model, not official): 1. Gaseous cycles have a major atmosphere or ocean reservoir and sedimentary cycles are centred in crust, soil or sediment; this is a reservoir classification, not a claim that every transfer is fast or reversible. 2. Rock weathering releases phosphate for biological uptake and later sedimentation; the owner treats phosphorus as having no significant atmospheric phase, so runoff and dispersed losses cannot be analysed as an atmospheric cycle. 3. Fossil-fuel combustion alters carbon transfers, synthetic fertiliser alters reactive nitrogen inputs, and phosphate mining and runoff alter phosphorus movement; different causes and reservoirs require different policy levers.

Q7. Distinguish NavIC from GAGAN in architecture, purpose and strategic value.

[10 marks | 150 words]

Model answer (129 words; practice model, not official): 1. NavIC or IRNSS is India's independent regional position, navigation and timing constellation, not India's name for GPS and not a global constellation. 2. The nominal NavIC architecture uses geostationary and inclined geosynchronous satellites to cover India and the surrounding region; nominal design must not be quoted as current functional health. 3. GAGAN is a Satellite Based Augmentation System that improves GPS accuracy and integrity for aviation; it is not an independent navigation constellation and does not replace NavIC. 4. Reference stations measure GNSS errors, master control computes corrections and integrity, uplink stations send messages to geostationary payloads, and compatible aircraft receivers use the rebroadcast. 5. Accuracy concerns closeness to the true position, while integrity concerns timely warning that a signal should not be used; safety-critical aviation needs both.

Q8. Explain the prevention-to-prosecution counter-terror chain and the institutions at each stage.

[10 marks | 150 words]

Model answer (101 words; practice model, not official): 1. The response chain separates prevention and intelligence, containment and specialised response, investigation by NIA or State police, prosecution in competent courts, finance disruption, and recovery or resilience. 2. The NIA Act, 2008 creates a central investigating agency for scheduled offences with concurrent jurisdiction; NIA does not replace the local police's first response, evidence preservation or ordinary policing role. 3. MAC and SMAC are intelligence-sharing and coordination platforms, not arresting, investigating or prosecuting commands. 4. Local police ordinarily reach an attack first and secure the scene, witnesses, evidence and public order, making State capacity structurally irreplaceable even in nationally investigated cases.

Q9. Explain the role of community institutions in disaster risk reduction.

[10 marks | 150 words]

Model answer (107 words; practice model, not official): 1. Community-based disaster risk reduction is participatory risk assessment, planning and action with affected communities, not transfer of the State's duty to untrained residents. 2. Local knowledge can identify routes, seasonal patterns, social networks and hidden vulnerabilities, but it should be combined with scientific and administrative information. 3. Communities often act before external services arrive; effectiveness depends on assigned roles, training, equipment, safety and coordination. 4. Panchayats and urban local bodies connect household-level risk information, local plans, shelters, volunteers, services and grievance redress to the DDMA. 5. Self-help groups, community-based organisations, youth groups, civil society and local leaders can support warning relay, evacuation, relief verification and recovery.

Q10. Distinguish hedging, speculation and arbitrage in derivatives markets.

[10 marks | 150 words]

Model answer (146 words; practice model, not official): Hedging uses a derivative to offset a pre-existing exposure. An exporter expecting dollar receipts may sell dollars forward, accepting basis and timing risk in exchange for more predictable rupee cash flow. Speculation creates or enlarges exposure: a trader buying index options without an offsetting portfolio seeks profit from price movement. Arbitrage combines offsetting positions to exploit inconsistent prices across related claims. The contract name does not reveal purpose. A future can hedge inventory or create a leveraged directional bet. Position size, maturity and underlying match determine whether total risk falls. Over-hedging can reverse exposure, while funding costs and execution delays make arbitrage imperfect. Regulation should therefore support genuine risk transfer through transparent prices, margins and reporting while controlling leverage, manipulation and unsuitable retail participation. Derivatives are neither inherently protective nor inherently harmful; their economic effect follows the exposure they create after considering the user's original position.

Q11. Examine the significance and limitations of India's 2022-23-base GDP series.

[15 marks | 250 words]

Model answer (219 words; practice model, not official): The 2022-23-base series, released by MoSPI on 27 February 2026, updates the benchmark used to measure India's changing economy. Its significance lies in five improvements. First, it separates activities within multi-activity companies rather than classifying the whole enterprise by one dominant activity. Second, annual survey data improve unincorporated-sector coverage. Third, GST and other administrative data widen corroboration. Fourth, agriculture and manufacturing use double deflation, while other sectors use stated volume or single-extrapolation methods. Fifth, Supply and Use Table integration and Proportional Denton quarterly benchmarking improve reconciliation and temporal consistency. MoSPI's 31 August 2026 NAS publication also records the incorporation of rebased PPI, IIP and Banking Services Price Index inputs, which revised estimates from 2022-23 onward. This illustrates why a base revision is more than changing one year's prices: sources, classifications, weights and methods can all change. Limitations remain. Administrative filings require validation; granular deflators demand reliable output and input prices; informal estimates still combine benchmarks with indicators; and methodological change creates a series break. February Second Advance Estimates, June Provisional Estimates and August NAS revisions are not interchangeable vintages. Therefore the revision should neither be accepted uncritically nor dismissed politically. Credibility depends on published concordances, revision studies, back series, accessible methodology and continuing survey improvement. Rebasing can improve relevance, but only transparent bridges preserve historical comparability and public trust.

Q12. Assess the achievements and limitations of India's Green Revolution.

[15 marks | 250 words]

Model answer (203 words; practice model, not official): India's Green Revolution was an institutional package: semi-dwarf cereal varieties were combined with assured irrigation, fertiliser, credit, extension, seed systems and price-procurement support. This complementarity, rather than seed alone, drove the breakthrough from the mid-1960s. Its principal achievement was a sustained rise in wheat and later rice yields and output. Larger marketed surplus strengthened domestic food availability, reduced chronic import dependence and expanded the policy capacity for public food management. It also stimulated rural demand for pumps, fertiliser, machinery, transport and processing. ICAR, IARI, State agricultural universities, extension workers and farmers created an adaptive research-and-diffusion system. However, the first gains were concentrated in irrigated Punjab, Haryana and western Uttar Pradesh and in wheat-rice systems. Rainfed regions, pulses and oilseeds benefited less initially. Unequal access to water, finance and machinery created distributional differences, while mechanisation altered labour demand unevenly. Ecological costs accumulated through groundwater depletion, nutrient imbalance, salinity, pesticide pressure, residue burning and monoculture. Assured infrastructure and market incentives then created crop path dependence. The balanced conclusion is not to reject productivity science but to redirect it. An evergreen-revolution strategy should combine climate-resilient varieties, balanced nutrients, water budgeting, diversified rotations, extension and viable alternative markets, preserving food security while reducing ecological and regional concentration.

Q13. Do you think India will meet 50 percent of its energy needs from renewable energy by 2030? Justify your answer. How will the shift of subsidies from fossil fuels to renewables help achieve the above objective? Explain.

[15 marks | 250 words]

Model answer (155 words; practice model, not official): The target must be stated correctly: India's COP26 commitment concerned about 50% of cumulative installed electric-power capacity from non-fossil sources by 2030, not half of total primary-energy consumption. India had reached 51.93% non-fossil installed capacity by 31 December 2025, while non-fossil generation was 30.41% in April-December 2025. Capacity therefore does not settle reliable energy contribution. Shifting support can improve relative prices by reducing general fossil consumption or producer support, financing renewable auctions, grids, storage, efficiency and targeted access, and internalising pollution and import risk. Abrupt withdrawal of lifeline electricity or clean-cooking support would be regressive; coal regions, workers and legacy debt need transition assistance. Renewable subsidies can also cause land, manufacturing or fiscal distortions. The appropriate shift is from general price suppression and unpriced externalities toward transparent, competitive support for generation, transmission, storage, demand flexibility and vulnerable users. Success should be measured by generation, curtailment, system cost, reliability and emissions, not capacity alone.

Q14. Assess the rationale and governance challenges of CPSEs.

[15 marks | 250 words]

Model answer (210 words; practice model, not official): Central Public Sector Enterprises can address long-gestation investment, strategic security, natural-monopoly features, universal service, regional development and missing private capability. Post-Independence CPSEs built heavy industry, energy, infrastructure and finance where domestic capital and technology were limited. Ownership alone, however, does not establish public value. Commercial firms need profitability, capital efficiency, innovation and customer discipline. Strategic or social mandates should be explicit, costed and separately monitored; otherwise losses can be defended as public service and managerial inefficiency becomes invisible. Repeated rescue may create a soft budget constraint, weakening investment and exit discipline. Governance is complicated by multiple principals. Administrative ministries articulate policy, DPE frames enterprise policy, DIPAM handles investment and asset transactions, boards manage companies, and CAG, Parliament, Companies Act, SEBI or sector regulators provide specified oversight. Informal ministerial intervention can dilute board accountability. Ratna status grants greater autonomy to qualifying CPSE boards. Current Maharatna, Navratna and Miniratna criteria combine prior classification, multi-year performance, scale or net-worth tests; autonomy still requires review and transparent capital allocation. Reform should classify each enterprise by strategic mandate and market structure, separate owner from regulator, appoint capable independent boards, publish public-service obligations, harden budget constraints and compare performance with relevant peers. Retention, merger, strategic sale, subsidiarisation or closure should follow evidence rather than blanket ideology.

Q15. Critically examine India's FRBM framework as a balance between fiscal discipline and flexibility.

[15 marks | 250 words]

Model answer (225 words; practice model, not official): The Fiscal Responsibility and Budget Management Act, 2003 seeks medium-term discipline, transparency and inter-generational equity. Section 3 requires fiscal-policy statements covering rolling targets, macro assumptions, strategy and expenditure commitments. Section 4 contains statutory reference targets, while sections 6 and 7 require transparency and explanation of deviations. Rules improve credibility by constraining deficit bias, exposing assumptions and anchoring debt expectations. They can lower uncertainty and preserve space for future shocks. Yet rigid annual ceilings may become pro-cyclical: abrupt consolidation during recession can weaken demand, employment and even revenue. Numerical compliance can also be manufactured through optimistic estimates, deferred payments or off-budget borrowing. The amended escape clause is therefore important. It permits deviation on enumerated grounds including national security, calamity, severe agricultural collapse, structural reform with unanticipated fiscal effects and a specified fall in real output growth. Annual deviation is capped at 0.5 percentage point of GDP, and reasons plus a return path must be laid before Parliament. The current position must be dated. Union Budget 2026-27 uses a declining Central-debt path, estimating debt at 55.6 per cent of GDP and fiscal deficit at 4.3 per cent in BE 2026-27. It also withholds the next two rolling-year targets because of global uncertainty. India therefore needs rule-bound flexibility: comprehensive coverage, realistic assumptions, protected productive expenditure, transparent escape use, independent scrutiny and a credible correction path.

Q16. Distinguish Green India Mission from CAMPA and classify the official GIM targets correctly.

[15 marks | 250 words]

Model answer (166 words; practice model, not official): 1. The Green India Mission is one of the original eight missions under the National Action Plan on Climate Change and combines adaptation and mitigation through forest and ecosystem restoration; it is not the CAMPA statute. 2. The retrieved MoEFCC page publishes mission targets of increasing forest or tree cover over 5 million hectares, improving quality over another 5 million hectares and enhancing forest-based livelihood income for about 3 million families; these are targets, not verified achievements. 3. The official page emphasises biodiversity, water, biomass, mangroves, wetlands, critical habitats, carbon storage, agroforestry, social forestry and urban or peri-urban tree cover; GIM is broader than compensating one diverted parcel. 4. MoEFCC describes convergence with CAMPA and MGNREGS and a role for local communities; convergence is a planning approach, not automatic pooling of every fund or proof of implementation. 5. CAMPA is diversion-triggered statutory compensation finance, whereas GIM is a broader mission with ecosystem-service objectives and multiple implementation channels; overlap in activity does not make them one programme.

Q17. Assess PFBR as a Stage-2 milestone without overstating deployment.

[15 marks | 250 words]

Model answer (162 words; practice model, not official): 1. Criticality means a self-sustaining controlled chain reaction with effective neutron multiplication near unity; first criticality is not grid synchronisation, rated power or commercial operation. 2. Thermal reactors use moderated slower neutrons, whereas fast breeder reactors deliberately avoid a moderator and retain a fast-neutron spectrum; coolant choice does not itself make a reactor thermal or fast. 3. Stage 2 centres on fast breeder reactors using plutonium-bearing fuel and fertile blankets to expand fissile resources; one prototype milestone is not an operational breeder fleet. 4. The 500 MWe PFBR at Kalpakkam was designed by IGCAR, built and commissioned by BHAVINI, and attained first criticality on 6 April 2026; the fetched DAE source did not establish grid or commercial operation. 5. IGCAR is the fast-reactor and sodium-technology R&D and design centre, while BHAVINI is the public-sector project execution and operating company for PFBR. 6. Installed megawatt capacity, first criticality, grid synchronisation, commercial operation and electricity generated over time are different measures and milestones.

Q18. Analyse the multidimensional challenges posed by external state and non-state actors to India's internal security.

[15 marks | 250 words]

Model answer (166 words; practice model, not official): 1. Internal security concerns peace, law and order, rule of law and sovereignty within India's territory; external security addresses foreign aggression, but neighbourhood linkages make the two domains inter-related. 2. Kautilya's four-fold classification distinguishes internal, external, internally aided external and externally aided internal threats; India's threat environment can combine all four. 3. A threat is a hostile actor or event, a vulnerability is an exploitable weakness, capability is the State's actual preventive or responsive means, and consequence is the realised harm when the first exploits the second. 4. External state and non-state actor analysis should identify actor, means, objective and intended end-state; proxy support can give a sponsor deniability without making sponsor and proxy the same legal actor. 5. Because policing is primarily State-owned while intelligence, border guarding and several central capabilities are Union-linked, Centre-State coordination is a structural constitutional requirement. 6. A law, notification, deployment, arrest, charge-sheet, agreement, implementation step and verified outcome are distinct evidentiary rungs and must never be collapsed in the analysis.

Q19. Explain the significance and limits of the Disaster Management (Amendment) Act, 2025.

[15 marks | 250 words]

Model answer (151 words; practice model, not official): 1. The Disaster Management (Amendment) Act, 2025 received assent on 29 March 2025 and came into force on 9 April 2025; the current position is not the unamended 2005 text. 2. Sections 8A and 8B inserted in 2025 give statutory status to the National Crisis Management Committee and High Level Committee. 3. Section 41A enables a State Government to constitute an Urban Disaster Management Authority by notification; the statutory permission does not prove constitution in every eligible city. 4. Section 44A enables a State Government to constitute a specialist State Disaster Response Force; this force must not be confused with the SDRF fund. 5. The amended architecture assigns National and State Plan preparation to NDMA and SDMA, while executive committees retain implementation and monitoring roles. 6. A statute, authority, plan, force, fund, notification or database mandate proves its legal or institutional rung only; operational readiness and disaster outcomes require separate evidence.

Q20. Examine the welfare and distributional effects of inflation through real wages, interest rates, debt and indexation.

[15 marks | 250 words]

Model answer (213 words; practice model, not official): Inflation changes welfare by altering real purchasing power and the value of nominal contracts. A worker whose money wage rises more slowly than CPI suffers a real-wage loss. The same applies to pensions and fixed incomes, with a sharper burden on poorer households because food and fuel occupy larger budget shares and cash forms a greater share of liquid wealth. The Fisher relation shows the interest channel: the ex ante real rate is approximately the nominal rate minus expected inflation. Unexpected inflation lowers the real value of fixed nominal repayment, benefiting a fixed-rate borrower and hurting the lender; unexpected deflation reverses the transfer. This result is not universal because floating rates reprice and borrowers may lose employment or income during tightening. Inflation tax is the erosion of real non-interest-bearing money balances. Governments may receive seigniorage or nominal tax gains, but also face higher indexed expenditure, welfare pressure and borrowing costs. CPI-IW-linked dearness allowance protects many covered employees and pensioners, demonstrating the value of indexation. Yet informal workers often lack such protection, and every indexed contract retains basis and adjustment-lag risk. Thus inflation is distributional, not merely aggregate. Policy should anchor expectations, protect vulnerable households through targeted support and avoid assuming that one national CPI matches every household's basket or that all borrowers gain.

END OF DETAILED SOLUTIONS